

mBERT vs MuRIL for Hinglish / Hindi-English Harmful Speech Detection

One-page research summary for review and project defense

Research Question

Does Indian-language-specific pretraining in MuRIL improve hate/offensive speech detection for Hinglish and Hindi-English code-mixed text compared with general multilingual pretraining in mBERT? Current evidence says the answer is conditional: the model with the higher observed score changes with dataset situation.

Datasets

Dataset	Rows	Positive label	Situation
Kaggle Hinglish Hate	4,780	hate	controlled Hinglish subset from Kaggle
CM Code-mixed	3,900	offensive	Indian politics / Twitter/X, hate-adjacent
THAR Religion	11,549	AntiReligion	YouTube targeted religious hate

Methods

- Models: mBERT and MuRIL fine-tuned as binary classifiers.
- Matched setting: train and evaluate on the same dataset situation.
- Robustness setting: evaluate trained checkpoints across other datasets and mixed-training conditions.
- Primary metric: Macro F1; positive recall is tracked because false negatives are missed harmful examples.

Main Matched Multi-Seed Result

Dataset	Higher observed	mBERT Macro F1	MuRIL Macro F1	Meaning
Kaggle	mBERT	67.5 +/- 2.1	58.1 +/- 5.7	mBERT has clear observed advantage
CM	mBERT narrow	77.7 +/- 1.9	76.1 +/- 2.3	both competitive; labels are offense-like
THAR	MuRIL	74.7 +/- 0.1	76.5 +/- 1.3	MuRIL has higher observed score

Current Interpretation

The central evidence is that model ranking changes by dataset: mBERT has higher observed scores on Kaggle and a small observed advantage on CM, while MuRIL has higher observed scores on THAR. Cross-dataset robustness is weak, so hate, offensive, and AntiReligion labels must not be treated as interchangeable. The experiments do not isolate which dataset factor causes each difference.

Limitations

- Mixed-training and cross-dataset transformer results are mostly single-seed evidence.
- Label definitions differ across datasets, which limits universal model-superiority claims.
- Kaggle metadata lists MIT; CM and THAR have no GitHub-detected license, so raw CM/THAR text should not be redistributed without further permission review.
- Reported matched scores are selection-set scores because best epoch was chosen using the same evaluation split.
- Small CM/THAR gaps are observed differences under this setup, not statistically established superiority.

Feedback Requested

Is the conditional claim clear enough? Are the dataset caveats acceptable? Which result should be prioritized for the next multi-seed run: mixed all-three training, Kaggle+THAR, CM+THAR, or cross-dataset transfer?